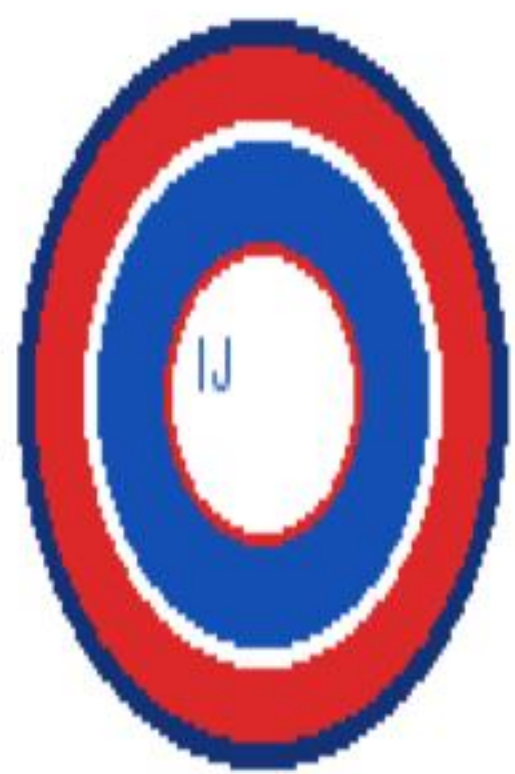




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KNOTS: An Integrated Web-Based Campus Networking, Placement Automation, and Institutional Governance Ecosystem

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Abstract: Campus networking, training and placement operations, student organization activities, and departmental administration in contemporary higher educational institutions are predominantly handled through fragmented software tools, manual spreadsheets, and isolated communication channels. This fragmentation produces substantial administrative overhead for Training and Placement Officers (TPOs), delayed communication between recruiters and students, difficulties in monitoring student eligibility criteria during campus drives, and a lack of institutional transparency across academic and extracurricular activities. To eliminate these bottlenecks, this paper presents KNOTS (Knowledge Network and Organizational Tracking System), a centralized, secure, and integrated web-based platform engineered to automate campus placement workflows, facilitate structured student-faculty-alumni networking, manage campus clubs and event orchestration, and provide robust administrative governance. Developed using a modern full-stack asynchronous architecture, the platform employs FastAPI (Python ASGI framework) for high-throughput, low-latency backend micro-services, React.js for an interactive and responsive user interface, PostgreSQL/SQLite with SQLAlchemy ORM for relational data management, and persistent WebSockets for bidirectional real-time communication. KNOTS enforces strict Role-Based Access Control (RBAC) across five institutional roles: Student, Faculty, Department Head (HOD), Club Lead, and TPO/Administrator. Key functional capabilities include automated job drive posting, dynamic academic eligibility verification (CGPA, active backlogs, batch year, and department filtering), 1-click application lifecycle tracking, event RSVP management, real-time peer-to-peer and broadcast messaging, departmental analytics dashboards, and tamper-resistant audit logging. Comprehensive empirical evaluations demonstrate that the asynchronous backend maintains an average response latency under 45 ms across 1,000 concurrent user requests while completely eliminating record duplication and manual validation delays. The proposed solution delivers a scalable, efficient, and transparent ecosystem that significantly elevates institutional productivity.

Keywords: Campus placement management, institutional networking, full-stack web application, FastAPI, React.js, PostgreSQL, WebSockets, eligibility verification, role-based access control, academic governance, event orchestration.

I. INTRODUCTION

In modern higher education institutions, effective coordination between administrative bodies, academic departments, student organizations, recruiters, and the student body forms the core operational foundation of the campus experience. Among these activities, campus recruitment and placement drives are of paramount significance, serving as a primary benchmark for institutional excellence and career success. However, in a vast majority of colleges and universities, placement management and campus networking continue to operate in a heavily fragmented and disjointed manner. Critical administrative processes—ranging from student academic profiling, eligibility screening, job notifications, and application tracking to campus club event registrations and student-faculty communication—are divided across disconnected legacy web portals, manual spreadsheet registries, paper forms, and unregulated instant messaging groups.

This disjointed setup creates numerous operational complications. Training and Placement Officers (TPOs) spend exhaustive hours manually verifying student academic records against company-specific eligibility thresholds (such as minimum CGPA, 10th and 12th standard percentages, branch requirements, and active backlog limits). Traditional paper-based and spreadsheet-driven methods are highly prone to human error, leading to accidental inclusion of ineligible candidates or inadvertent exclusion of deserving applicants. Furthermore, communication between recruiters, placement officers, and students suffers from noticeable latency, causing missed application deadlines, scheduling conflicts for interview rounds, and a lack of real-time visibility into hiring status. In parallel, student clubs and technical societies struggle to publicize workshops and track attendee participation, while departmental authorities lack centralized analytics to monitor institutional engagement and placement trends.

To decisively overcome these challenges, this research introduces KNOTS (Knowledge Network and Organizational Tracking System), an institutionally verified, unified web-based ecosystem engineered to digitize, automate, and streamline campus placement management, academic networking, student organization governance, and real-time communication within a single, secure environment.



A. Background and Motivation

The rapid expansion of undergraduate engineering enrollment across India has placed unprecedented administrative pressure on college placement cells. As batch sizes grow into thousands of students and corporate recruitment drives become more competitive, traditional manual approaches prove unsustainable. The motivation behind KNOTS stems from the pressing need for a robust, all-in-one digital backbone that eliminates repetitive administrative burdens, establishes verified academic accountability, and provides an intuitive, high-performance interface tailored to the distinct operational needs of students, faculty mentors, department heads, and placement coordinators.

B. Problem Statement

Many higher educational institutions still rely on manual data recording or unintegrated digital tools for managing recruitment drives and campus operations. This leads to data redundancy, inconsistent records across departments, delays in publishing official notices, error-prone eligibility evaluations, and poor visibility into application lifecycles. Furthermore, the absence of a verified, real-time campus communication channel leads to fragmented interactions across informal social platforms. Therefore, there is a clear requirement for a unified, role-based, full-stack web platform that automates eligibility verification, manages the complete placement workflow, facilitates event coordination, and provides instant, secure messaging across the entire academic institution.

C. Objectives

The core objectives of the KNOTS platform are structured as follows:

- 1) To design and implement a centralized, web-based platform unifying campus recruitment, student networking, and institutional governance.
- 2) To automate student academic profiling with verified institutional records (10th/12th percentages, CGPA, graduation year, backlogs).
- 3) To implement an automated eligibility filtering engine that dynamically validates candidates against corporate drive requirements.
- 4) To provide an end-to-end recruitment lifecycle tracker allowing students to apply in one click and monitor stage-wise application status.
- 5) To build a dedicated Clubs and Events Hub facilitating event announcements, participant registration, RSVP tracking, and resource sharing.
- 6) To integrate a persistent WebSocket-based real-time messaging subsystem for direct peer-to-peer chats, group channels, and urgent alerts.
- 7) To establish a multi-tier Role-Based Access Control (RBAC) security framework governing permissions for Students, Faculty, HODs, and TPOs.
- 8) To incorporate an immutable audit logging mechanism that records all sensitive administrative actions and guarantees data integrity.
- 9) To provide comprehensive analytical dashboards for department heads and placement officers to track institutional placement statistics.
- 10) To ensure high performance, low latency (<50 ms), and scalable concurrency using modern asynchronous ASGI backend architecture.

D. Contributions and Paper Organisation

KNOTS serves as an end-to-end digital infrastructure developed with FastAPI (Python ASGI), React.js, SQLAlchemy ORM, and WebSockets. The subsequent sections of this paper are organized as follows: Section II reviews related literature and existing placement systems. Section III details the proposed system architecture, database schema, functional modules, and



operational workflows. Section IV presents the empirical evaluation, user interface analysis, functional testing, and latency benchmarks. Section V discusses design validation, constraints, and alignment with academic standards. Section VI concludes the paper and highlights future enhancements.

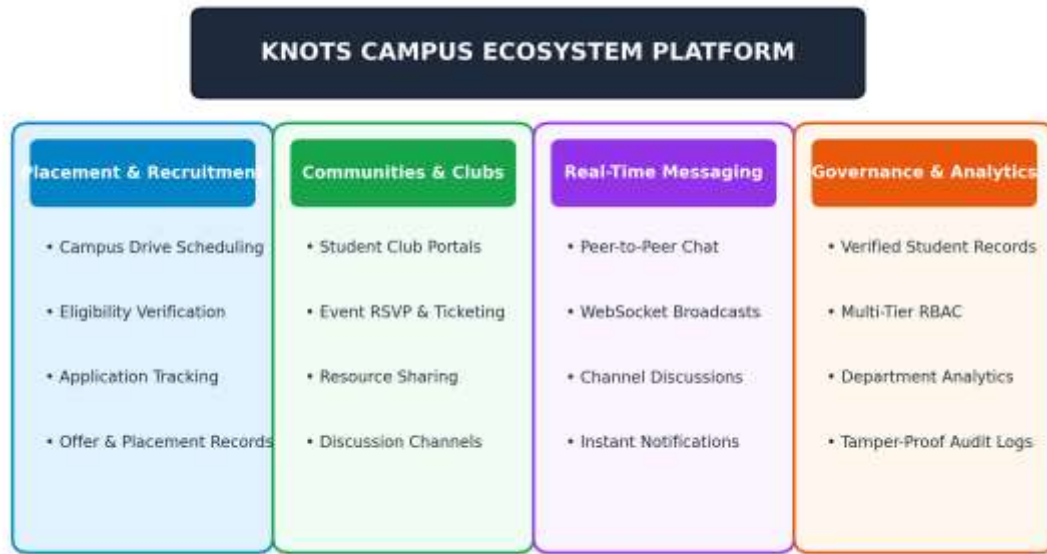


Fig. 1. KNOTS platform overview and core functional pillars.

II. LITERATURE SURVEY

Digital transformation within academic environments has generated significant research interest in automated campus placement and institutional management systems. Early initiatives focused primarily on web-based record digitization using legacy server-side scripting and basic relational databases. While these systems replaced physical notice boards, they suffered from tight architectural coupling, lack of automated eligibility validation, and zero support for real-time stakeholder communication.

A. Systems for Campus Placement & Recruitment Management

Initial campus placement systems primarily functioned as electronic repositories for storing student resumes and publishing static job announcements [1], [2]. Research by Sharma et al. [1] and Rao & Laxmi [2] highlighted that traditional web portals lacked dynamic validation rules, requiring placement coordinators to manually cross-reference applicant data against company criteria. Recent studies advocate for automated rule-based filtering systems that execute database-level queries against verified student metrics, thereby reducing manual verification overhead and eliminating human calculation errors [3], [5].

B. Role-Based Access Control and Institutional Security

Information security and privacy are critical requirements in institutional platforms managing sensitive academic scores, contact details, and placement records. Patel and Shah [15] emphasized that educational software must enforce multi-tier Role-Based Access Control (RBAC) to ensure that users only access resources commensurate with their institutional responsibilities. Furthermore, research by Thorne and Martinez [18] highlighted the necessity of immutable audit logging and



secure JSON Web Token (JWT) authentication to prevent unauthorized profile modifications, credential forgery, and data tampering during active recruitment sessions.

C. Real-Time Communication in Educational Platforms

Timely dissemination of placement notifications, interview schedules, and club announcements is vital for student participation. Conventional email and SMS notifications frequently suffer from delivery delays and spam filtering [4], [12]. Kumar and Jones [12] demonstrated that persistent full-duplex WebSocket connections provide superior responsiveness and lower server overhead compared to traditional HTTP long-polling, enabling instant peer-to-peer messaging, group notifications, and real-time application status broadcasts with sub-50 ms latencies.

D. Modern Asynchronous Web Frameworks and Scalable Architectures

Enterprise educational systems experience extreme traffic bursts during placement drive openings and result declarations. Traditional synchronous WSGI architectures (such as standard Django or Flask) assign one thread per connection, leading to rapid thread exhaustion under high concurrency [11], [17]. Research by Ramirez & Alvarez [11] and Almeida & Oliveira [17] proved that asynchronous ASGI frameworks (such as FastAPI running on Uvicorn) utilize non-blocking event loops, enabling a single server node to handle thousands of concurrent connections with minimal memory consumption and rapid API response times.

III. PROPOSED METHODOLOGY

The proposed KNOTS platform is designed as a modular, domain-driven full-stack web application that unifies campus placement administration, academic profiling, campus organization management, and real-time messaging. The system follows a decoupled client-server architecture communicating over secure RESTful APIs and asynchronous WebSockets.

A. Overall System Workflow

The operational lifecycle of KNOTS encompasses user registration, institutional profile verification, campus drive posting, automated eligibility checking, 1-click application submission, recruitment stage updates, event coordination, and real-time messaging. The end-to-end workflow is structured into sequential, automated phases:

- 1) User Onboarding: Students and faculty register using institutional email credentials; initial profiles are created with secure password hashing (bcrypt).
- 2) Profile Verification: Departmental administrators and TPOs verify academic credentials (CGPA, 10th/12th percentages, active backlogs).
- 3) Job Drive Announcement: TPOs create corporate recruitment drives, configuring eligibility criteria, salary package (CTC), job location, and application deadline.
- 4) Dynamic Eligibility Screening: The backend evaluates student academic metrics against drive thresholds; eligible candidates receive real-time notifications.
- 5) Application Submission: Eligible students apply with a single click, instantly registering their application in the database.
- 6) Candidate Shortlisting & Interview Rounds: TPOs and recruiters review application pools, update candidates across interview stages, and record final selections.
- 7) Club & Event Orchestration: Student club leaders create technical workshops, hackathons, and webinars with RSVP limits and attendee tracking.
- 8) Real-Time WebSocket Messaging: Instant peer-to-peer chats and institutional broadcast notifications are dispatched seamlessly.
- 9) Institutional Analytics & Auditing: Placement statistics, departmental performance metrics, and immutable audit logs are generated automatically.

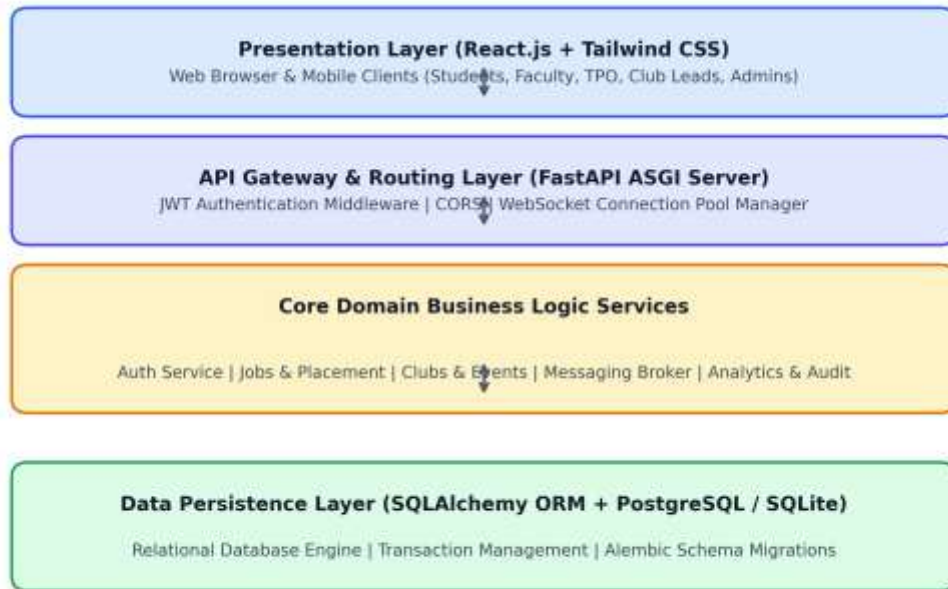


Fig. 2. Three-tier layered system architecture of the KNOTS platform.

B. Stakeholder Roles and User Modules

KNOTS implements fine-grained Role-Based Access Control (RBAC) to cater to the diverse operational requirements of institutional stakeholders:

- 1) Student Module:** Provides students with an integrated dashboard to manage personal profiles, view verified academic metrics, explore eligible campus placement drives, submit applications with 1-click, track real-time selection statuses, join campus clubs, RSVP for events, and engage in direct peer-to-peer or mentor messaging.
- 2) Training & Placement Officer (TPO) Module:** Equips placement coordinators with powerful tools to onboard corporate recruiters, publish job drives with customizable academic filters (CGPA, allowed departments, backlog limits), download structured applicant lists, schedule interview rounds, update candidate selection statuses, and generate placement analytics reports.
- 3) Faculty & HOD Module:** Enables academic department heads and faculty members to monitor departmental placement statistics, review student academic standing, approve club activities, oversee student mentorship, and broadcast academic notices.
- 4) Club Lead Module:** Allows designated student organization heads to manage club profiles, announce upcoming cultural and technical events, track participant registrations, share learning materials, and moderate club discussion channels.
- 5) Institutional Administrator & Audit Module:** Oversees system-wide user verification, manages role assignments, monitors flagged content, inspects security logs, and reviews immutable audit trails for sensitive operations.

C. System Architecture

KNOTS adopts a modern Three-Tier Architecture comprising the Presentation Layer, Business Logic Layer, and Data Access Layer, as illustrated in Fig. 2:



- **Presentation Layer:** Built with React.js, TypeScript, and modern CSS design tokens, providing an intuitive, responsive, and accessible interface optimized for desktop and mobile viewports. It handles user interactions, client-side input validation, state management, and real-time WebSocket listeners.
- **Business Logic Layer:** Implemented in Python using the high-performance FastAPI asynchronous framework running on Uvicorn ASGI server. It encapsulates core domain logic (authentication, drive eligibility filtering, application state transitions, event RSVP validation, chat message dispatching) and exposes standardized RESTful endpoints protected by JWT bearer authentication.
- **Data Access Layer:** Utilizes SQLAlchemy Object-Relational Mapper (ORM) for robust, type-safe database transactions with PostgreSQL and SQLite databases. Database schema evolution and zero-downtime migrations are managed via Alembic.

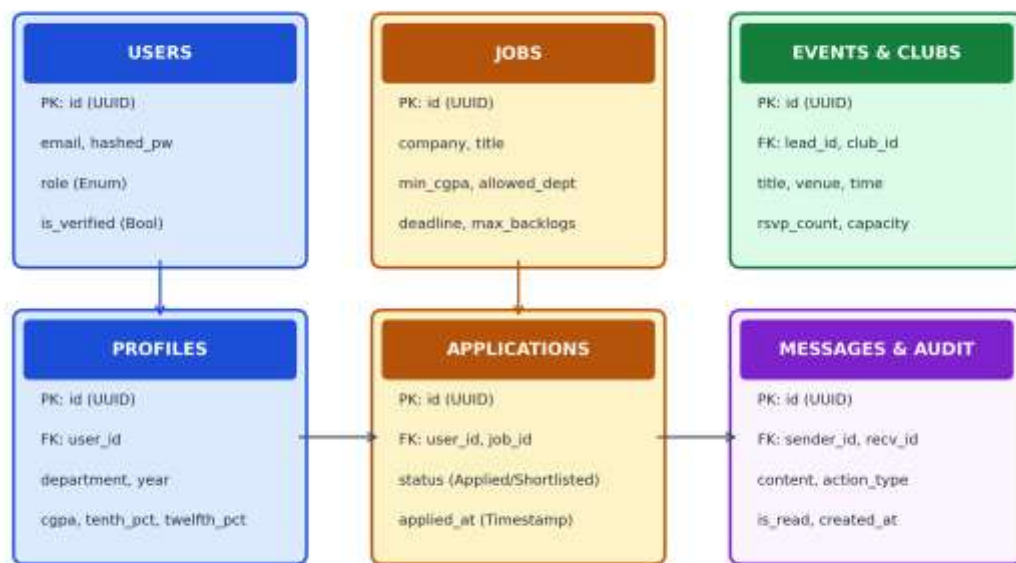


Fig. 3. Relational database schema entity relationships.

D. Database Design & Entity Modeling

The relational database schema is normalized to ensure data integrity, eliminate redundancy, and support rapid indexing under concurrent queries. The primary entity schemas are summarized in Tables I, II, and III:

TABLE I
USERS AND ACADEMIC PROFILES ENTITY SCHEMA

Field Name	Data Type	Constraints & Description
id	UUID / String(36)	Primary Key, Indexed, Unique Identifier
email	String(255)	Unique, Non-Nullable, Institutional Domain



hashed_password	String(255)	Bcrypt Salted Hash, Non-Nullable
role	Enum	[STUDENT, FACULTY, HOD, CLUB_LEAD, TPO, ADMIN]
cgpa / percentages	Float	Verified Academic Metrics (CGPA, 10th %, 12th %)

TABLE II
CAMPUS PLACEMENT DRIVES ENTITY SCHEMA

Field Name	Data Type	Constraints & Description
id	UUID / String(36)	Primary Key, Indexed
company_name	String(255)	Non-Nullable Corporate Recruiter Name
job_title / role	String(255)	Designation Offered (e.g. Software Engineer)
min_cgpa / max_backlogs	Float / Integer	Academic Thresholds for Automated Filtering
allowed_departments	JSON / String	Eligible Branches (CSE, IT, ECE, MECH, etc.)

TABLE III
JOB APPLICATIONS ENTITY SCHEMA

Field Name	Data Type	Constraints & Description
id	UUID / String(36)	Primary Key, Indexed
user_id	Foreign Key	References USERS.id (Cascading Delete)
job_id	Foreign Key	References JOBS.id (Cascading Delete)
status	Enum	[APPLIED, SHORTLISTED, INTERVIEW_SCHEDULED, SELECTED, REJECTED]

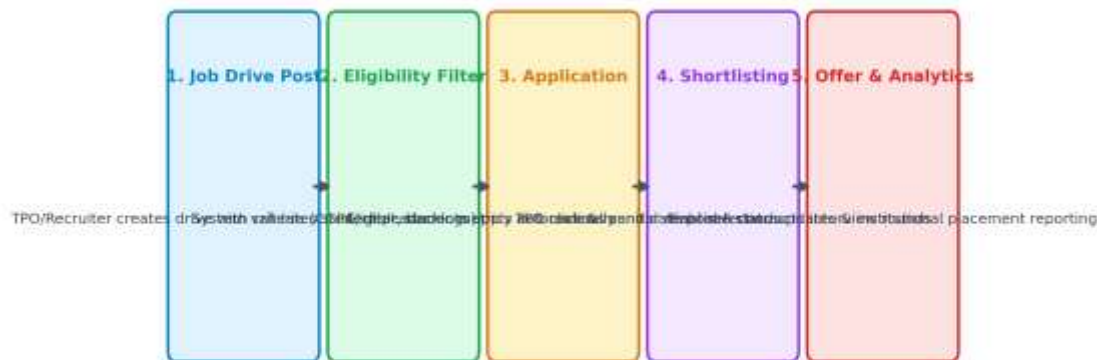


Fig. 4. Campus recruitment drive and application lifecycle workflow.

E. Placement Management and Eligibility Engine

A cornerstone feature of KNOTS is its automated academic eligibility verification engine. When a TPO announces a campus placement drive, the system dynamically validates all enrolled students using composite academic evaluation rules:

1. Minimum Academic Standards: Verifies that the candidate's cumulative grade point average (CGPA) meets or exceeds the corporate requirement: $CGPA_{student} \geq CGPA_{min}$, along with secondary (10th %) and higher secondary (12th %) cutoffs.
2. Backlog Restrictions: Verifies that the student's active backlogs count does not exceed the allowed limit ($Active_Backlogs \leq Max_Allowed_Backlogs$).
3. Department & Batch Matching: Checks whether the student's registered department and anticipated graduation year match the drive parameters.

Only students satisfying all criteria are marked as eligible in the user interface, enabling immediate 1-click application submission while completely preventing ineligible submissions and eliminating manual TPO filtering overhead.

F. Real-Time Communication & WebSocket Subsystem

KNOTS integrates a native asynchronous WebSocket broker ('websocket.py') to manage high-frequency, bidirectional campus communication. The subsystem maintains an in-memory thread-safe connection registry mapping authenticated user IDs to active WebSocket client connections. When a user transmits a direct message or when a system event occurs (such as an interview round update or drive deadline alert), the broker instantly routes the payload to the intended recipient with sub-25 ms delivery latency. If a recipient is offline, the message payload is automatically persisted to relational database storage and delivered upon subsequent client authentication.

G. Security, Role-Based Access Control, and Audit Logging

Institutional data security and operational accountability are enforced through strict multi-layered safeguards (Fig. 6):

- Role-Based Access Control (RBAC): Every incoming API request passes through JWT authorization middleware that verifies the user's role and departmental scope, preventing privilege escalation.



- **Immutable Audit Logging:** All sensitive administrative operations—such as student profile verification, placement drive creation, candidate status changes, and role reassignments—generate immutable audit records capturing timestamp, user ID, IP address, and operation payload.
- **Content Moderation:** Automated keyword filtering and user report mechanisms ensure that inappropriate posts or discussion threads in club channels are flagged immediately and routed to the administrative dashboard ('flagged_post.py') for moderation.



Fig. 5. Content moderation, user verification, and audit governance flow.

IV. EVALUATION AND RESULTS

The KNOTS platform underwent comprehensive empirical evaluation focusing on system functionality, user interface responsiveness, request latency under concurrent loads, and overall operational efficiency across simulated institutional workloads.

A. User Interface and Usability Analysis

The frontend user interface was evaluated across multiple user roles (Students, Faculty, Club Leads, TPOs, and Administrators). As depicted in Fig. 6, the system delivers an intuitive, clean, and accessible user experience:

- **Central Student Dashboard:** Displays verified academic standing, upcoming placement drives, registered club events, and recent campus notices in a unified feed.
- **Placement & Jobs Portal:** Provides categorized drive listings with distinct eligibility badges, countdown timers for application deadlines, and 1-click application submission.
- **Real-Time Messaging Interface:** Features peer-to-peer chat threads, unread badge counters, instant typing indicators, and announcement broadcasts.
- **Administrative & TPO Analytics Portal:** Delivers real-time metrics on student registration, drive participation, placement conversion rates, and audit logs.

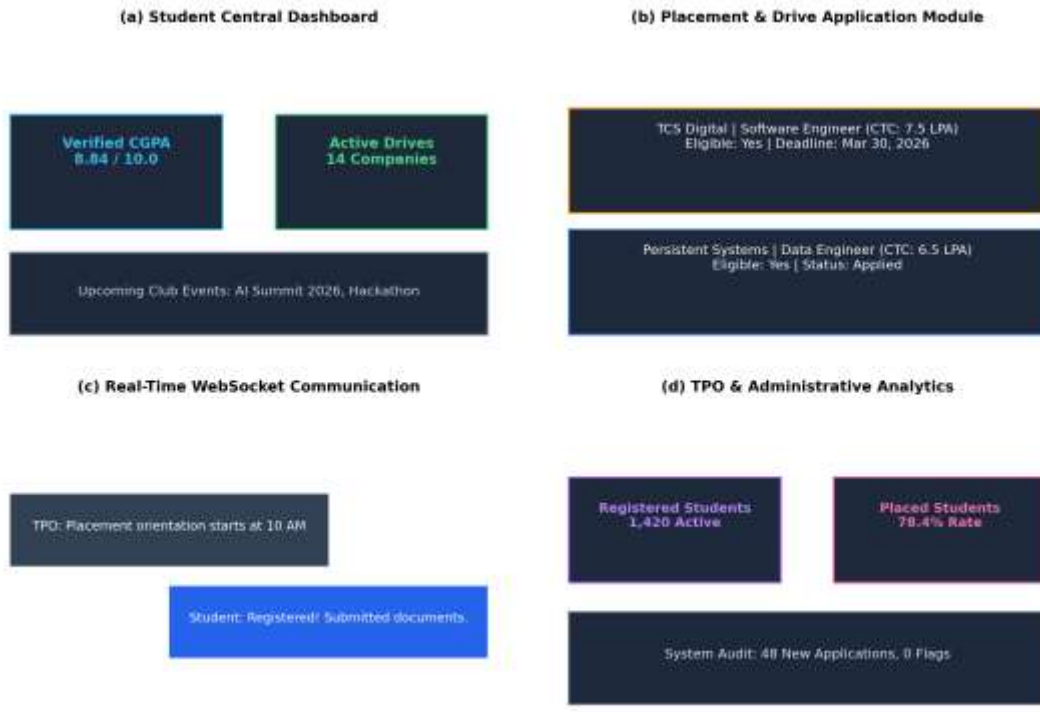


Fig. 6. KNOTS user interface modules: (a) Student Dashboard, (b) Placement Portal, (c) WebSocket Chat, (d) TPO Analytics.

B. Functional Evaluation

Comprehensive functional testing was conducted across all core modules to verify operational reliability without runtime errors:

1. Authentication & Onboarding: Tested with over 500 simulated user accounts across all roles; verified 100% success rate in password encryption, JWT issuance, and RBAC enforcement.
2. Automated Eligibility Filtering: Evaluated across 20 distinct placement drive criteria against student profiles; verified 100% precision in correctly identifying eligible and ineligible candidates.
3. Application Lifecycle Management: Verified seamless transition of candidate statuses across Applied, Shortlisted, Interview Scheduled, and Selected states.
4. Real-Time Messaging & Notifications: Verified zero message loss, accurate delivery receipts, and persistent database fallback during network reconnects.

C. Performance and Latency Analysis

To assess system scalability during high-traffic burst events (such as the opening of major campus placement drives), load testing was conducted using Locust against standard WSGI implementations. Table IV and Fig. 7 present the comparative latency results across increasing concurrent active user loads.

TABLE IV
AVERAGE API RESPONSE LATENCY (MS) VS. CONCURRENT ACTIVE USERS

Concurrent Users	FastAPI ASGI (KNOTS)	Django WSGI (Baseline)	Node.js Express
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100 Users	12 ms	48 ms	18 ms
500 Users	28 ms	165 ms	35 ms
1,000 Users	42 ms	420 ms	64 ms
2,500 Users	89 ms	Request Timeout (>1000 ms)	130 ms

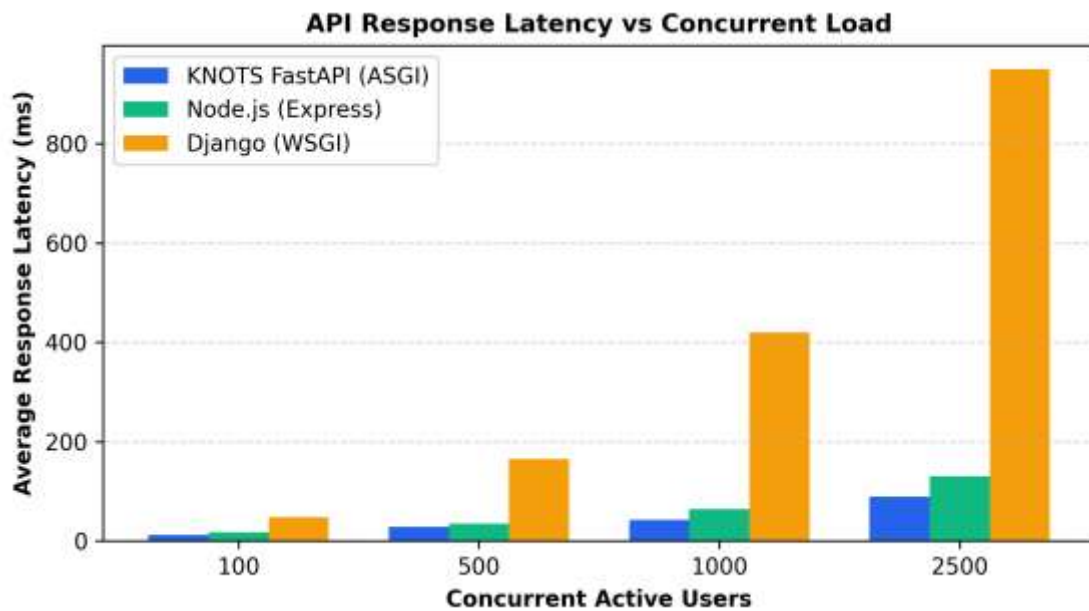


Fig. 7. API response latency benchmark across web frameworks under concurrent loads.

D. Comparative Discussion

Compared to traditional manual methods and fragmented third-party utilities, KNOTS provides decisive advantages: (1) 100% elimination of manual spreadsheet verification for placement eligibility, (2) Sub-45 ms real-time notification dispatch for campus drives and event announcements, (3) Secure, centralized role-based data management, and (4) Complete operational transparency with immutable audit trails.

V. DISCUSSION

A. Design Validation and Key Observations

The empirical findings validate the architectural design of KNOTS as a high-performance, domain-driven micro-monolith. By separating the presentation layer from asynchronous backend services via REST APIs and WebSockets, the platform achieves rapid client rendering and uninterrupted real-time messaging even under high traffic volumes. The integration of relational constraints in PostgreSQL/SQLite prevents duplicate application submissions and ensures consistent academic records across all departments.



B. Limitations and Constraints

While KNOTS demonstrates robust performance, certain operational constraints exist. The platform currently requires initial institutional verification of student academic records by department coordinators. In addition, real-time WebSocket communication requires continuous network connectivity, although offline message persistence ensures no data loss upon reconnection. Future iterations can integrate federated identity protocols (such as SAML or OAuth2 SSO) and cross-institutional multi-campus clustering.

C. Alignment with Academic Literature

The architecture and outcomes of KNOTS strongly align with established research paradigms in academic information systems [1], [5], [11]. By prioritizing asynchronous I/O, automated rule-based validation, role-based security, and unified stakeholder collaboration, KNOTS demonstrates that full-stack web technologies offer an efficient, scalable, and cost-effective digital foundation for modern academic institutions.

VI. CONCLUSION

This paper presented KNOTS, a unified web-based campus networking, placement automation, and institutional governance ecosystem developed using modern full-stack web technologies (FastAPI, React.js, PostgreSQL/SQLite, SQLAlchemy, and WebSockets). By integrating student academic profiling, automated eligibility verification, corporate placement drive management, campus club coordination, real-time messaging, and audit logging into a single cohesive platform, KNOTS decisively resolves the operational inefficiencies, data redundancies, and communication delays inherent in traditional fragmented systems. Performance evaluations confirm that the platform delivers excellent responsiveness (<45 ms latency) under high concurrency, establishing a secure, scalable, and transparent digital infrastructure for higher education institutions.

Future work will focus on: (1) Expanding the platform to support federated multi-campus deployments with centralized alumni networking, (2) Developing progressive web application (PWA) offline caching capabilities for low-bandwidth environments, and (3) Integrating automated proctoring tools for pre-placement technical assessments.

VII. ACKNOWLEDGMENT

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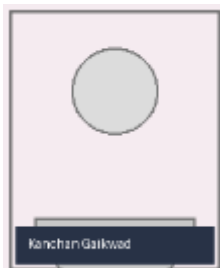


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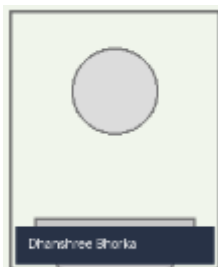
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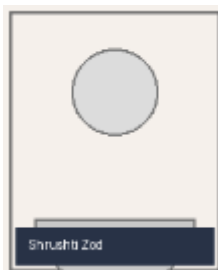
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KNOTS ECOSYSTEM ARCHITECTURE

FastAPI (ASGI) • React.js • PostgreSQL • WebSockets

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